Contact

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www.linkedin.com/in/deepak-lingaraju-a4702b275 (LinkedIn)

Top Skills

Signal Processing
XAI
Feature Engineering

Languages

English (Full Professional)
Kannada (Native or Bilingual)
German (Professional Working)

Certifications

Python for Data Science and Machine Learning Bootcamp
Optimization Onramp
Introduction to Statistical Methods with MATLAB
Reinforcement Learning Onramp
Machine Learning with MATLAB

Deepak Lingaraju

Computer Vision enthusiast pursuing Mechatronics Engineering Masters Degree | Deep Learning | AI | Python.
Germany

Summary

I'm passionate about developing AI-driven systems for automation, robotics, and intelligent manufacturing, integrating machine learning, deep learning, and computer vision with real-world engineering challenges.

My interests include robot perception, sensor fusion, and autonomous decision-making, with hands-on experience working in ROS 2 environments that bridge deep learning and computer vision for intelligent robotic behavior.

I'm continuously expanding my expertise in Machine Learning, ROS 2 and C++, aiming to create adaptive and efficient AI systems for next-generation autonomous and industrial applications.

Experience

Volkswagen Group

Machine Learning & Deep Learning Research Intern
September 2025 - May 2026 (9 months)
Wolfsburg

• Developed machine learning and hybrid deep learning models (CNNs, RNNs) for quality prediction and process analysis in resistance spot welding (RSW) using large-scale industrial sensor time-series data

• Applied signal processing and feature engineering techniques (FFT, wavelet analysis, statistical feature extraction) to identify process patterns and build predictive models for nugget diameter estimation and weld quality assessment under varying process disturbances.

• Implemented domain adaptation (e.g., DANN) and Explainable AI (XAI) approaches to improve model robustness, interpretability, and trust in industrial manufacturing environments.

• Built scalable machine learning pipelines using Python and AWS SageMaker for model training, evaluation, and experimentation, while collaborating with manufacturing and materials engineering teams to translate data-driven insights into practical process improvements.

• Focused on developing interpretable AI models for multivariate time-series manufacturing data, exploring hybrid ML–DL approaches and feature attribution techniques, with experiments and tools documented in GitHub repositories for reproducibility and collaboration.

Kannegiesser
Master Thesis
June 2024 - December 2024 (7 months)
Vlotho, North Rhine-Westphalia, Germany

• Built a custom convolutional backbone and integrated it into a YOLOv3–v4–inspired detection model tailored for garment region identification in automated processing
• Designed a single-scale detection architecture optimized for fixed camera–garment distance and analyzed performance across multiple feature map depths for small vs. large objects.
• Developed a custom YOLO-based loss function and conducted comparative experiments using transfer learning to benchmark model performance.

Incresol Engineering Solutions Private Limited
Mechanical Design Engineer
November 2019 - April 2020 (6 months)
Bengaluru, Karnataka, India

Education

University of Duisburg-Essen
Master of Engineering, Mechatronics · (September 2021 - September 2024)

Visvesvaraya Technological University
Bachelor's degree, Mechanical Engineering · (June 2015 - June 2019)

Sri Bhagawan Mahaveer Jain College
Pre-University PCME, Physics, Chemistry, Mathematics and Electronics · (June 2013 - June 2015)